

The Neuro Reset Plan

Brian, this plan is more than just switching medications—it's about optimizing your brain chemistry, restoring balance after years of stress and alcohol exposure, and building a stronger foundation for mental resilience.

By making these strategic adjustments, we're not just managing symptoms—we're rewiring your system to work with you, not against you. This means:

- ✓ Less panic, more clarity.
- ✓ Steady energy, fewer spikes and crashes.
- ✓ Gut-brain alignment for a better mood.
- ✓ The right medication for **YOUR metabolism.**

This isn't just another medication change—it's a precision-tuned reset designed for long-term emotional stability and cognitive sharpness.

You've done the hard part by recognizing the patterns. Now, let's upgrade the system.

We'll monitor progress and adjust as needed. I'll be with you every step of the way. Let's make this work for you. 💡

- Anna Tran, PMHNP-BC

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### Predicted Pharmacogenetic & Pharmacodynamic Metabolism Phenotype for Brian

Based on Brian's medication response, history of alcohol use, anxiety sensitivity, and reported side effects, we can infer some likely pharmacogenetic and pharmacodynamic traits affecting his medication metabolism and neurotransmitter balance.

#### 1. CYP2C19 & CYP2D6 – Poor Metabolizer of Citalopram & Wellbutrin

- Citalopram (Celexa) metabolism is heavily dependent on CYP2C19. If Brian is a poor metabolizer (PM), citalopram builds up in his system, increasing QT prolongation risk, restlessness, and paradoxical anxiety.

- Wellbutrin (Bupropion) is metabolized by CYP2B6 and inhibits CYP2D6, leading to higher propranolol levels and worsening autonomic instability.

- If he is a CYP2D6 poor metabolizer, he may have reduced dopamine clearance, leading to increased anxiety sensitivity and an exaggerated response to norepinephrine stimulants.

✅ Phenotype Prediction:

🚨 CYP2C19 Poor or Intermediate Metabolizer → Higher risk of citalopram buildup, side effects, and cardiac risks.

🚨 CYP2D6 Intermediate to Poor Metabolizer → Wellbutrin inhibits his metabolism further, amplifying anxiety and increasing propranolol levels.

🚨 CYP2B6 Slow Metabolizer → Extended Wellbutrin half-life, worsening overstimulation.

## 2. COMT “Worrier” Gene – Slow Dopamine Clearance → Anxiety Sensitivity

- If Brian has the Met/Met variant of the COMT gene, his brain breaks down dopamine slowly, making him:

- More prone to rumination and emotional overthinking.
- More sensitive to stimulant effects (Wellbutrin increasing dopamine & norepinephrine).
- More reactive to stress and environmental stimuli.

✅ Phenotype Prediction:

🚨 COMT “Worrier” (Met/Met) Genotype → Stronger anxiety response to dopamine fluctuations. Wellbutrin likely increased agitation rather than improving motivation.

## 3. Brain-Gut Axis Dysfunction – Likely Linked to Alcohol Use & PPIs

- Long-term alcohol use disrupts gut microbiota, reducing serotonin and GABA production, making anxiety harder to regulate.

- Prolonged omeprazole use has likely reduced his B12, folate, and vitamin D absorption, leading to neurotransmitter deficiencies that worsen depression and anxiety.

- His sensitivity to medications suggests heightened autonomic reactivity, possibly due to low GABA and dopamine depletion from past alcohol use.

✅ Phenotype Prediction:

🔴 Gut-Brain Axis Dysfunction → Low serotonin and GABA levels due to long-term alcohol use & PPI-induced nutrient depletion.

🔴 Reduced B12 & Folate Absorption → Decreased serotonin synthesis, contributing to residual depression.

### Summary of Brian's Likely Pharmacogenetic & Pharmacodynamic Traits

| Trait                       | Prediction                     | Impact on Medication Response                                            |
|-----------------------------|--------------------------------|--------------------------------------------------------------------------|
| CYP2C19                     | Poor/Intermediate Metabolizer  | Citalopram builds up → increased QT prolongation, activation, anxiety    |
| CYP2D6                      | Poor/Intermediate Metabolizer  | Wellbutrin slows metabolism → increased norepinephrine → overstimulation |
| CYP2B6                      | Slow Metabolizer               | Wellbutrin lasts longer → prolonged stimulation & anxiety                |
| COMT "Worrier" (Met/Met)    | Increased dopamine sensitivity | → greater panic susceptibility                                           |
| Gut-Brain DysfunctionLikely | Reduced serotonin & GABA       | → higher anxiety baseline                                                |

### Why Duloxetine Is the Right Choice Based on His Pharmacogenetics

✓ Balanced SNRI approach → Less reliance on single pathways (serotonin & norepinephrine support without excessive dopamine shifts).

✓ Avoids QT prolongation risk of citalopram → Safer cardiac profile.

✓ More predictable metabolism compared to Wellbutrin → Fewer overstimulation issues.

✓ Helps correct gut-brain neurotransmitter depletion by improving norepinephrine & serotonin balance.

### Next Steps

- ♦ Consider pharmacogenetic testing (IDX) to confirm these predictions. This can be done to in my office
- ♦ Monitor B12, folate, and vitamin D levels to optimize neurotransmitter production.
- ♦ Observe anxiety response to Duloxetine 30–60 mg and adjust accordingly.

Conclusion: Based on Brian's symptoms, history, and likely metabolism, he appears to be a CYP2C19 poor metabolizer, CYP2D6 intermediate-poor metabolizer, and a COMT "Worrier" genotype. This explains why citalopram and Wellbutrin caused excessive anxiety and why duloxetine is a better fit.

We will monitor symptoms closely and fine-tune the plan as needed. 🚀